

科学真实性、准确性与完整性审查文档

对应稿件：*Storm-tide magnitude and standardized connected terrain yield non-interchangeable coastal screening priorities in a Northwest European station network*

审查定位：code-data-table-figure-manuscript 穿透式内部审计

自动约束状态：全部通过

1. 审查结论

本轮大修没有把审稿意见当作文字层面的“解释补丁”，而是重新冻结唯一主流程，并从原始/中间数据重算区域指标。当前 68 站主结果的算术和拓扑约束通过；主要 Spearman、Kendall、岸段内双侧置换、保持岸段结构的 top-k null、Moran's I 和 six-sector resampling 均直接来自当前 source tables。稿件已删除旧 degree-window 数值、15/20 m denominator sensitivity、未定义的 response categories，以及把 top-set mismatch 解释为 beyond-chance discordance 的写法。

现阶段可以确认的是：代码生成的当前表格与图件内部一致，结果支持“两个区域筛查排序不能互相替代”。不能确认、也没有声称的是：静态地形指标等于真实洪水风险，或者七个本地 DTM case 能代表全部 68 站。垂向基准未统一和缺少独立真实洪水基准仍是投稿前的主要外部证据限制。

2. 四个 Po 问题的处理证据

1. **DeltaDTM cutoff 与 15 m denominator：已重构。** 当前文件由 GeoTIFF tag 和 v1.1 README 识别为 DeltaDTM v1.1.1，cap 为 30 m EGM2008；2024 论文描述的是 v1.0 的 10 m + MSL 范围。主流程不再使用 15 m 或 20 m lowland denominator，仅用有限 class-0 represented land；正文明确这种比例不是 nominal 10 km square 的全部陆地比例。

2. **旧窗口数值混入 Supplement：已移除。** 新 Supplementary Information 只从 Primary_terrain_components_68stations.csv 和 Terrain_robustness_grid_74stations.csv 导出。旧 S2/S6、Fig5_* 和 degree-window 表不再构成稿件证据。

3. **sector/country 混用：已统一。** 地图、block resampling 和 leave-one-out 使用同一组六个 coastal-sector identifiers；country omission 不再被称为 sector sensitivity。

4. **未定义 response category：已删除。** 正文只报告连续值和绝对面积；Supplementary Figure S4 报告七个诊断站的连续差值和 IoU，不再报告“4/7 category changed”。

3. 产品版本与 mask 审计

product	release	tile_tag	vertical_reference	release_cap_m	paper_v1_o_cap_m	mask_outside_support_fill	official_255_cells_in_primary_analyses_windows	outside_support_cells_in_primary_analyses_windows	note
DeltaDTM	v1.1.1	DeltaDTM v1.1.1 / DeltaDTM.jl v1.1.1	EGM2008	30	10	254	0	3320	The 2024 paper documents v1.0; the downloaded v1.1 README documents the 30 m EGM2008 cap. Outside-support fill is kept distinct from class 255 and excluded from all land denominators.

旧代码曾同时用 255 表示 DeltaDTM 官方 clipped class 和 boundless/reprojection fill。新版代码将窗口外支持固定为 254。扫描得到的 72 个 source mask tiles 中未发现官方 255；旧窗口中出现的 255 因而不能解释为真实 clipped land。当前自动门禁要求 254/255 分离，并禁止 lake、outside support、clipped support 或任意 no-data 作为 marine seed。

4. 自动科学约束

Check	Pass	Observed	Criterion or meaning
Robustness-grid rows	True	2960	74 stations x 40 recorded configurations
Unique stations	True	74	regional candidate inventory
Primary stations	True	68	COAST-RP match <=6 km
Connected area <= below-threshold area	True	0	violations
Four-neighbour <= eight-neighbour	True	0	violations
Metric decomposition identity	True	4.26326e-14	maximum absolute percentage-point error
Official class 255 in primary windows	True	0	cells
Outside-support sentinel distinct	True	3320	class-254 cells recorded and excluded

这些检查证明当前输出满足代码自身声明的面积与拓扑关系。它们不证明输入产品没有系统误差，也不等同于独立洪水观测验证。

5. 统计证据边界

terrain_m etric	n_sites	spearman	kendall	permutatio n_two_sided_p	top_n	top_overla p	top_overla p_expecte d_independ ent	spearman _sector_b ootstrap_p 025	spearman _sector_b ootstrap_p 975	sector_adj usted_ran k_associat ion	sector_str atified_per mutation_ two_sided _p
connected _reference _land_pct	68	-0.333168	-0.210711	0.0063993 6	14	1	2.88235	-0.647188	0.178855	-0.491883	0.0003999 6
below_refe rence_lan d_pct	68	-0.309272	-0.187884	0.0085991 4	14	1	2.88235	-0.624297	0.190665	-0.45364	0.000899 91
conditiona l_connecti vity_pct	68	0.108145	0.0717274	0.378762	14	5	2.88235	-0.276457	0.32199	0.198325	0.420458
connected _area_km 2	68	-0.340955	-0.208077	0.0047995 2	14	1	2.88235	-0.626084	0.208442	-0.488862	0.0001999 8

station-label permutation 的显著性与 spatial inference 必须分开阅读。connected-land share 和 connected area 的 station-level p 较小，但 six-sector ranges 跨 o。Top-20% overlap 为 1/14，独立随机期望为 2.88; exact lower-tail p = 0.152。因此稿件只写 limited recovery / weak concordance / non-interchangeability，不写 unusual mismatch、distortion、mis-rank 或 proven complementarity。

主推断采用保持岸段组成的空间零模型：RP10 与 connected-land share 的 top-20% overlap 为 1/14，空间零模型 95% 包络为 [0, 4]，lower-tail p=0.257。观察值处在包络内，因此该结果只证明两个清单恢复度有限，不证明错配异常大于空间随机。

指标分解还说明 composite connected share 的负关联主要来自 below-2 m land prevalence; conditional connectivity 的 rho 约为 o。这一结果直接纠正了把低 composite fraction 一概解释为“低连通性”的构念混淆。

分母几何与拓扑

terrain_metric	terrain_metric_label	n_sites	raw_spearman_water_vs_metric	partial_spearman_controlling_represented_land_fraction	spearman_metric_vs_represented_land_fraction	spearman_water_vs_represented_land_fraction
below_reference_land_pct	lowland_prevalence	68	-0.309272	-0.310448	-0.0976448	0.0031683
conditional_connectivity_pct	conditional_connectivity	68	0.104533	0.105806	-0.127793	0.0031683
connected_reference_land_pct	connected_lowland_share	68	-0.333168	-0.335911	-0.137191	0.0031683
connected_area_km2	connected_area	68	-0.340955	-0.341566	0.0511509	0.0031683
connected_window_pct	connected_window_share	68	-0.340955	-0.341566	0.0511509	0.0031683

terrain_metric	n_sites	rank_spearman_4n_vs_8n	top20_n	top20_overlap	membership_changes	maximum_eight_minus_four	median_eight_minus_four
connected_reference_land_pct	68	0.961102	14	11	6	44.0569	0.00732171
connected_area_km2	68	0.966828	14	11	6	23.5498	0.0038732
conditional_connectivity_pct	68	0.900571	14	13	2	91.5269	0.581332

RP10 与 represented-land fraction 的 rho 接近 0，控制该比例前后主关联几乎不变。4 邻域和 8 邻域在区域排名上高度一致，但 top-20% 仍有站点成员变化，证明站点优先级对拓扑定义比区域总体结论更敏感。

COAST-RP 匹配审计

68 个纳入站点的最大匹配距离为 2.811 km；下列 6 个站点因超过 6 km 规则被排除，而不是在出图后人工删除：

station	recomputed_match_dist_km	coast_rp_source_lon	coast_rp_source_lat	match_exclusion_reason
Cordemais France	22.6474	-2.19	47.278	excluded_nearest_coast_rp_point_gt6km
Le Pellerin France	28.388	-2.014	47.014	excluded_nearest_coast_rp_point_gt6km
Nantes Usine Brulee	39.423	-2.014	47.014	excluded_nearest_coast_rp_point_gt6km
Donges France	8.27799	-2.19	47.278	excluded_nearest_coast_rp_point_gt6km
Montoir De Bretagne	6.89698	-2.19	47.278	excluded_nearest_coast_rp_point_gt6km
Paimboeuf France	14.3371	-2.19	47.278	excluded_nearest_coast_rp_point_gt6km

6. 本地 DTM product-and-datum cross-check

station	delta_connected_2 m_reference_land_ pct	local_connected_2 m_reference_land_ pct	delta_connected_2 m_area_km2	local_connected_2 m_area_km2	connected_pixel_io u_after_reprojectio n	local_dtm_vertical_ datum_note
Sheerness	42.7002	9.62959	20.3643	4.61207	0.214623	Ordnance Datum Newlyn / EA product native datum
Newlyn	0.44611	0.416648	0.276293	0.259144	0.527548	Ordnance Datum Newlyn / EA product native datum
Lowestoft Po24 Uk	1.41616	6.16525	0.669507	2.92848	0.110977	Ordnance Datum Newlyn / EA product native datum
Immingham Po26 Uk	0.405551	0.078999	0.24946	0.0487939	0.0417246	Ordnance Datum Newlyn / EA product native datum
Denhelder Hel Nl	71.7296	4.36493	20.8545	1.27433	0.0554507	Normaal Amsterdams Peil / AHN product native datum
Delfzijl Del Nl	96.7041	89.7843	61.3031	57.1869	0.925796	Normaal Amsterdams Peil / AHN product native datum
Hoek van Holland	2.63966	1.43861	1.41098	0.771845	0.217064	Normaal Amsterdams Peil / AHN product native datum

Supplementary Figure S4 的 IoU target grid、grid shape、transform、mask resampling 和文件 SHA-256 全部记录在 `TableS_local_dtm_product_datum_crosscheck.csv` 。 Environment Agency 和 AHN 数据未与 EGM2008 统一，因此本文不使用 validation 一词，也不把差异归因给 DeltaDTM 单一误差。七站为 diagnostic cases，不把变化比例外推到 68 站总体。

7. 图件真实性与视觉检查

Figure	Pixels	DPI target	Source data	Visual QA
Fig1_process_terrain_design.png	3542 x 2249	500	Primary_station_coastal_sect tors.csv; Primary_sample_flow.csv	inspected at original resolution; no deliberate cell enlargement
Fig2_regional_screening_agr eement.png	3394 x 3360	500	Primary_terrain_component s_68stations.csv; Return_period_sensitivity.cs v; Spatial_sector_bootstrap_int ervals.csv; Topk_spatial_null_curves.cs v	inspected at original resolution; no deliberate cell enlargement
Fig3_connected_terrain_con trasts.png	3669 x 3934	500	Fig3_predefined_site_selecti on.csv; Fig3_threshold_response_so urce.csv	inspected at original resolution; no deliberate cell enlargement
Fig4_metric_decomposition _robustness.png	3710 x 3318	500	Primary_terrain_component s_68stations.csv; Denominator_adjusted_asso ciations.csv; Topology_sensitivity_summa ry.csv; Rank_robustness_one_at_a _time.csv	inspected at original resolution; no deliberate cell enlargement
FigS1_water_level_definition .png	3443 x 1456	500	Fig2_water_level_divergence _with_uncertainty.csv; TableS_tidal_regime_metad ata_source_tracked.csv	inspected at original resolution; no deliberate cell enlargement
FigS2_process_coherence.pn g	5122 x 2354	500	Fig3_driver_correlations.csv; event-composite source tables	inspected at original resolution; no deliberate cell enlargement
FigS3_connectivity_gallery.p ng	3462 x 3415	500	Primary_terrain_component s_68stations.csv; frozen DeltaDTM windows	inspected at original resolution; no deliberate cell enlargement
FigS4_elevation_product_da tum_sensitivity.png	3574 x 2975	500	TableS_local_dtm_product_ datum_crosscheck.csv	inspected at original resolution; no deliberate cell enlargement
FigS5_coastrp_match_audit. png	3466 x 1590	500	Match_distance_audit_74sta tions.csv; Match_distance_histogram_ source.csv	inspected at original resolution; no deliberate cell enlargement

Figure 3 的三个站由机器可读规则选出。低值站点只对原始 connected cells 做局部放大，不改变主地图中的像元面积；high/low 视觉平衡通过指标分解和三条实算 threshold curves 实现，而不是调数值。Figure 4 取消离散类别，Figure 2 明确区分 raw station estimate、sector range 和 spatially constrained top-k null。

8. 关键文件 SHA-256

File	Bytes	SHA-256
data/raw/coast_rp/COAST-RP.zip	6798999	13242fe8b682230420357dbac202b10590f41dcoc9e378b9c2c69f91962704c9
data/raw/gssr/metadata/eraint.geojson	307194	aab77394a0bf87b3ae611131e26b489a7abe9dd67add071aeeeeoda4b3bab46e6
data/raw/deltatdm/index/README.md	2326	9da10b0576ff4368a756bd5646b091fab6b7c7c7d217a6f758fbe0803f826e00
data/raw/deltatdm/zips/mask_tiles.zip	428178062	3f8c6999f90346359e60cc6c757786d43fd1cb65e4ab13125faaa80606487775
data/figure_source/Primary_terrain_components_68stations.csv	31386	c516a97298a87ddd8d1746d8975e23afff94043a2a030c2e8eee6707c6c6893e
data/figure_source/Terrain_robustness_grid_74stations.csv	1321599	5708031c3492aa47f2bf456d52df5dcbe1d22ac427e5dc7e5d168723b56539af
data/figure_source/Rank_robustness_one_at_a_time.csv	35926	dc43ead84439e0d72c1f71ddf9bde1cca9334fdbf399e2385a020b3f3b5e2b2d
data/figure_source/Spatial_sector_bootstrap_intervals.csv	1316	2e73a799fcb280cb37daab1a6d7347b632c546b0a4a9bc4d70512d1055e93124
data/figure_source/Topk_spatial_null_curves.csv	5811	ea385515afb10a2448160909057daf2e4e6f2b5356f76d1d6349b9331ee16042
data/figure_source/Return_period_sensitivity.csv	8146	51397a192a4e7e413cof22d1afb044d17eaca7c956f234c09c00fc49629e8a3b
data/figure_source/Denominator_adjusted_associations.csv	884	0f8dea8b42895a42fa1d672edfe2a022dae900643d6d93c2316633de802a2823
data/figure_source/Topology_sensitivity_summary.csv	422	85f7263bc4da748e07eb96bf6231c94ffe4072d62aa8294743b91ea62916602b
data/figure_source/Match_distance_audit_74stations.csv	37578	d255f5fa00644be1d371c8dfia789c307bbe89f7d463842f151b04620338odfo
data/figure_source/TableS_local_dtm_product_datum_crosscheck.csv	9733	6b9c4656676341d87c2ea8b0a84c307c984ed1e04439dca7b3e0563329ae9928
scripts/add_mask_aware_terrain_diagnostics.py	32421	aa8e38aaf0eedfac3628eabd84270fcdbb2c279e1a1fbd8c1afbdb7ef1cf5e16
scripts/rebuild_frozen_primary_analysis.py	21686	057e8bbd4011cac63bfedc58a85c94981eaa1aaa4dd1726b87b7f4f5ca7389cc
scripts/add_external_review_diagnostics.py	28100	b9bb20e4a29c0961500d7961b4a1c7f8e70b70285750b5812e2742f03d312e21
scripts/add_local_dtm_validation.py	20251	f4320a331c77492aa95f1216bc6e12301a5d2ce76a49b46914cff3d636c8a3ef
scripts/make_major_revision_figures.py	41950	4e1c09d6362481d8b283ee1e40b848f52f16c47014381198938db2ece8c2c006
data/raw/local_dtm/delfzijl-delnl_local_dtm_10m.tif	2204083	a8b3589a24ca9dbcb00d18e24c916e193d37a9b73034c185bf29fcf80a9cbb35
data/raw/local_dtm/denhelder-helnl_local_dtm_10m.tif	1310648	b950e177da667509c85bb8a4dc77c4dcbac5b56d85507ea706e968568af15e79
data/raw/local_dtm/hoekvanholla-hvhnl_local_dtm_10m.tif	1848802	e960ac322ae8e703055a75739d6dee6a65c2ee367e9dcodd8af1cf6ff600563
data/raw/local_dtm/immingham-p026-uk_local_dtm_10m.tif	9437675	8567cb377b27480b51f3dbeef5501784f3eed237e0ec7e8058ff09789ed6e7

File	Bytes	SHA-256
data/raw/local_dtm/lowestoft-p024-uk_local_dtm_10m.tif	6291923	d95f97c91d4df295e6113b98fcbdb4d3bedac1a35c3751464e0b8e61ab2819
data/raw/local_dtm/newlyn-p001-uk_local_dtm_10m.tif	9437675	fef345c91868dd64e5a2e1118oad066b7a1f8ce92305cf26990d70fd0c9e51c2
data/raw/local_dtm/sheerness-p015-uk_local_dtm_10m.tif	9437675	40c81f2a969e0044184d5fb40ebc11728dca5376d2707386e29c90def027cadd

此清单用于确认后续 manuscript、report 或 publication package 是否仍由同一基线生成。大体积上游数据不应凭文件名认定相同，必须复核 hash 或上游 DOI/version。

9. 仍未消除的风险

Risk	Status	Evidence boundary
垂向基准统一	未完成	DeltaDTM/ODN/NAP 尚未转换到共同 local-MSL reference; Supplementary Figure S4 只能解释为 combined product/datum/provider-resampling sensitivity。
独立真实洪水基准	未完成	没有 observed or dynamic flood-hazard truth dataset, 因此不能声称 connected-terrain ranking 是真实优先级。
样本代表性	受限	68 站来自 GSSR metadata inventory, 且岸段分布不均; Denmark-Norway 仅 2 站。
本地 DTM 服务重采样	受限	WCS scaleFactor 生成 nominal 10 m; 连续高程插值核由服务端控制。
公开审稿仓库	已完成	代码、派生数据、图表源数据、文稿和审查证据已发布至 https://github.com/Coucou2016/20260522-coastal-flood-scientific-data ; 上游原始文件以来源和 SHA-256 清单追踪。正式归档 DOI 尚待版本冻结。
潮汐元数据	部分完成	正式 cross-product return-level difference 的潮汐语境限制在 13 个 source-verified sites; Brest 和 Hoek 仍不得进入正式 tidal inference。

10. 可验收的投稿前目标

- 共同垂向基准：**至少将 3 个英国和 3 个荷兰 case 的 terrain 与 coastal MSL 转换到明确共同参考；验收为方法给出 offset 来源、公式和不确定性，重算后的 source table/figure 可一键复现。若无法完成，则 Supplementary Figure S4 保持 sensitivity 定位。
- 独立结果基准：**在可匹配站点上引入 observed 或 dynamic hazard outcome；验收为预先定义评价指标并分别比较 water-only、terrain-only 和 combined screen。完成前不使用 predictive complementarity。

3. **独立站点框架**：用不依赖 GSSR inventory 的 coastal station/shoreline sample 重算主关系；验收为 sample-flow 记录全部候选、排除原因和 coastal-spacing/sector coverage。
4. **公开复现入口**：GitHub 公开入口已经建立；下一项验收是在独立临时目录中克隆仓库并运行不依赖上游大文件的合同测试，再记录 release tag、commit 和 archive hashes。正式投稿版本冻结后可再生成归档 DOI。
5. **投稿元数据**：补齐 authors、affiliations、funding、contributions；验收为 HTML、Markdown 和 PDF 中无“待补充”占位。

11. 复现入口

核心重算命令为 `python scripts/rebuild_frozen_primary_analysis.py`。本地 DTM 交叉检查为 `python scripts/add_local_dtm_validation.py`，主图为 `python scripts/make_major_revision_figures.py`，论文和补充材料分别由 `python scripts/build_standalone_paper.py` 与 `python scripts/build_supplementary_information.py` 生成。自动科学合同使用 `PYTEST_DISABLE_PLUGIN_AUTOLOAD=1 python -m pytest tests -q --import-mode=importlib`。